

Multimodal Item Scoring for Natural Language Recommendation via Gaussian Process Regression with LLM Relevance Judgments

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GPR-LLM, modeling **MULTIMODAL** item relevance from a few LLM-judged anchor, can improve NL Rec!

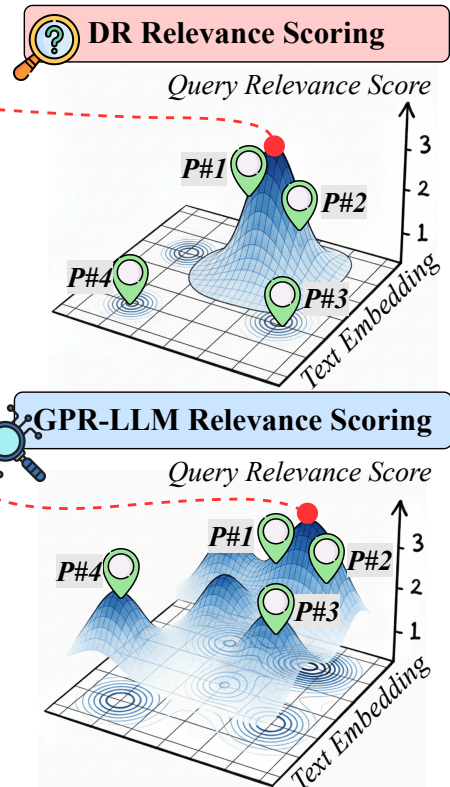
1. Why is this hard?

Item relevance in NL Rec is often multimodal, but dense retrieval (DR) is often unimodal

NL Request:
Family friendly cities for vacations

● Query
● Relevant Passages

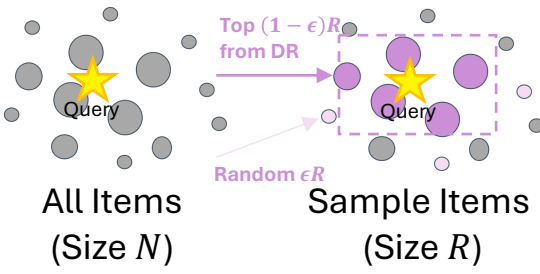
- Passage#1: A family friendly city...
- Passage#2: Kid-Friendly Attractions, Zoos, aquariums...
- Passage#3: Outdoor & Nature Activities; National parks...
- Passage#4: Healthcare & Emergency Readiness ...



2. Our idea: GPR-LLM

1 Sample

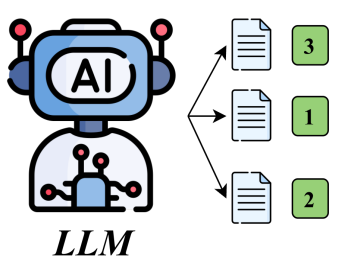
Sample $R \ll N$ Items from Top DR + ϵ random exploration



Design Choice for Sampling:
 ϵ : How much to explore (0 = greedy DR)

2 Label

Ask LLM to judge the sampled items



4 Predict

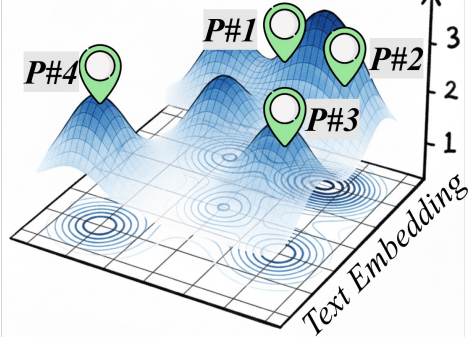
Use GP Posterior mean to score all items

#1	#2	#3	#4	...	#N
2.9	2.7	0.8	1.4		1.5

Design Choice for GP:
Kernel function: linear, cosine, RBF

3 Fit GP

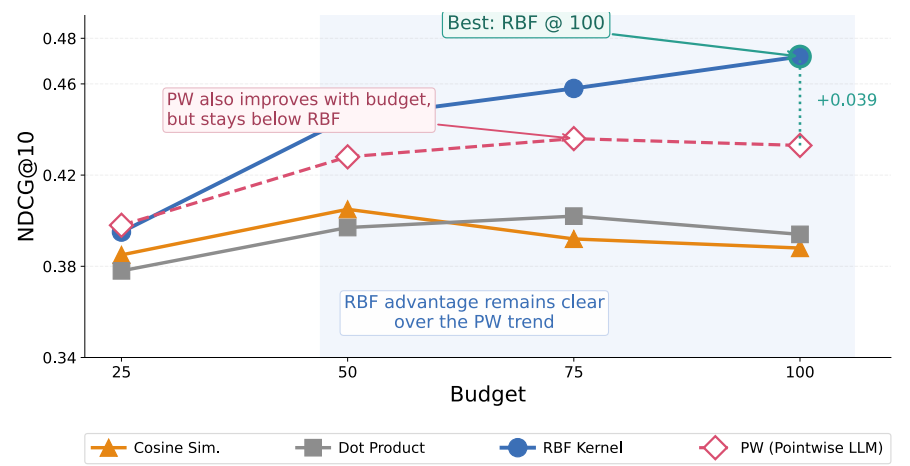
Fit a query specific Gaussian Process model



3. Does it work?

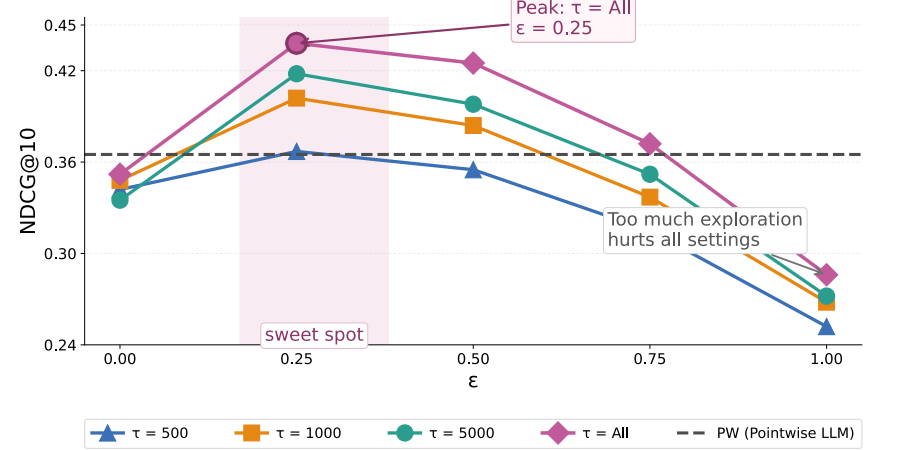
A Kernel Choice

GP + RBF > Pointwise LLM > GP + Cosine / Dot Product



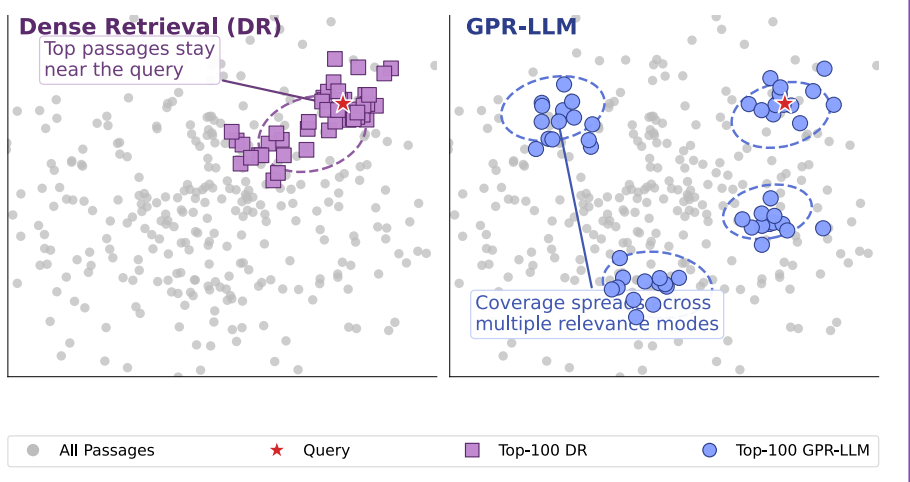
B Sample strategy

A little exploration finds better anchors.



C Why the gains?

GPR-LLM + RBF captures multimodal relevance.



Key quantitative findings

- Up to 65% improvement over pointwise LLM relevance scoring
- Often matches or beats baselines using 2x more labels
- Robust across 4 datasets and 3 LLMs